

Geometric G and Its Formal Substitution into Einstein's Equations

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Abstract

We present an empirical geometric parametrization of the gravitational constant G based on the golden ratio ϕ , the integer 31, and $\sqrt{5}$. When G is expressed in units derived from the Earth (meter from the meridian quadrant, kilogram from one liter of water), the length and mass units combine so that the result can be expressed with an effective factor of 10^3 , yielding a quantity with dimensions of inverse square time only:

$$G_{\text{Earth}} = \frac{31^2 \times \sqrt{5}}{\phi^{12}} \times 10^{-8} \text{ s}^{-2}.$$

The corresponding value in SI units is

$$G_{\text{Koch}} \approx 6.67354674496 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2},$$

which lies between the measurements of Schlamminger (2014) and CODATA 2018. Substituting this expression into Einstein's field equations formally replaces the coupling constant $8\pi G$ with a combination of geometric invariants. No modification of general relativity is proposed; the substitution is purely algebraic. The physical interpretation of this geometric form remains open.

1 A Formal Replacement of the Coupling Constant

Einstein's field equations read

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}.$$

Substituting the geometric expression for G (now in s^{-2}) gives

$$G_{\mu\nu} = 8\pi \left(\frac{31^2 \times \sqrt{5}}{\phi^{12}} \times 10^{-8} \right) T_{\mu\nu}.$$

This is a formal algebraic substitution. The equations retain their full mathematical structure. Whether the appearance of ϕ and 31 in the coupling constant may reflect a deeper geometric property of spacetime remains an open question.

2 Geometric Parametrization of G

From the identity (derived in [5])

$$\frac{\phi^{12}}{\sqrt{5}} = 144 + \frac{1}{720} + \mathcal{O}(10^{-8}),$$

we obtain

$$G_{\text{Koch}} = \frac{31^2 \times \sqrt{5}}{\phi^{12} \times 10^{11}} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}.$$

The numbers 31, 10^{11} (see [4]), and $1/720$ are introduced through independent geometric constraints described in [3, 5, 6].

3 Earth-Based Units: Dimensional Reduction

Let the Earth's meridian quarter be M . Define

$$1 \text{ m} = \frac{M}{10^7}, \quad 1 \text{ dm} = \frac{M}{10^8}.$$

The kilogram is historically tied to the mass of one liter (1 dm^3) of water, whose density is $\rho_{\text{water}} = 999.97 \text{ kg/m}^3$. Hence

$$1 \text{ kg} \approx \rho_{\text{water}} \times (1 \text{ dm})^3 \approx 1 \times \frac{M^3}{10^{24}}.$$

Now form the combination that appears in G :

$$\frac{1 \text{ m}^3}{1 \text{ kg}} \approx \frac{(M/10^7)^3}{M^3/10^{24}} = 10^{-21} \cdot 10^{24} = 10^3.$$

– From the meter definition we obtain 10^{-21} ; from the kilogram definition we obtain 10^{24} ; their product yields 10^3 . The small deviation due to $\rho_{\text{water}} = 999.97$ (relative 0.003%) is taken as an empirical tolerance.

Applying the same scaling to G_{Koch} :

$$G_{\text{Earth}} = G_{\text{Koch}} \times 10^3 = \frac{31^2 \times \sqrt{5}}{\phi^{12}} \times 10^{-8} \text{ s}^{-2}.$$

Thus in these Earth-based units the gravitational constant has dimension s^{-2} only. Note that $86\,400 = 120 \times 720$ with $720 = 6!$, and the same integer 720 appears in the golden-ratio identity above. The recurrence of 720 in two otherwise independent constructions is noted as a numerical coincidence requiring further justification. The origin of these numbers is discussed in [3, 5].

4 Two Independent Constraints for the Integer 31

The integer 31 serves as a discrete anchor in two separate geometric relations.

Quadratic closure: $31^2 = 961$, and the corresponding relation

$$G_{\text{Koch}} \times 10^{11} \times 144$$

matches this value with a relative gap of about 0.001%.

Scaling optimum: The continuous equation $(\phi^6 \times n)^4 = 10^{11}$ yields the continuous optimum $n \approx 31.338$. For the discrete integer $n = 31$, the uncorrected deviation is 4.25% (i.e., $(\phi^6 \times 31)^4$ is 4.25% below 10^{11}). After applying the micro-step correction $12/720 = 0.016667$ the residual reduces to 0.020% (see [4]).

These two constraints are presented as independent numerical criteria. No neighbouring integer (30 or 32) achieves comparably small residuals under both criteria.

Table 1: Comparison of neighbouring integers

n	ΔG (vs CODATA)	$\Delta(\phi^6 n)^4$ (vs 10^{11})	Note
30	6.358%	16.018%	too small
31	0.011%	4.25%	discrete anchor
32	6.544%	8.718%	too large
31.338	$\sim 0\%$	$\sim 0\%$	continuous optimum

5 What This Is and Is Not

This work is:

- an empirical geometric parametrization of G ,
- a demonstration that G can be numerically represented using ϕ , 31, and $\sqrt{5}$,
- a formal substitution into Einstein’s field equations.

This work is not:

- a derivation of gravity from first principles,
- a modification or replacement of general relativity,
- a claim that spacetime is discrete or spiral-structured,
- an attempt to explain the physical origin of ϕ , 31, or 720 beyond the references provided.

6 Conclusion

The gravitational constant can be represented in a compact geometric form involving only the golden ratio, the integer 31, and $\sqrt{5}$. When expressed in units derived from Earth's meridian and the mass of water, G reduces to a pure inverse square of time. The same integer 720 that defines the second also appears in the geometric residual of $\phi^{12}/\sqrt{5}$ (see Appendix). These observations may indicate a numerical connection between Earth's geometry, the golden ratio, and the measured value of G . The physical significance of this formulation remains an open question.

No experimental G values were used in the derivation; all numbers originate from independent geometric constraints documented in [3, 4, 5, 6].

A Derivation of the Key Identity via Binet's Formula

The Fibonacci numbers satisfy Binet's formula:

$$F_n = \frac{\phi^n - \psi^n}{\sqrt{5}}, \quad \psi = -\phi^{-1}.$$

For $n = 12$:

$$F_{12} = 144 = \frac{\phi^{12} - \phi^{-12}}{\sqrt{5}}.$$

Rearranging gives:

$$\frac{\phi^{12}}{\sqrt{5}} = 144 + \frac{\phi^{-12}}{\sqrt{5}}.$$

Numerically, $\phi^{-12} \approx 0.003105624$, so

$$\frac{\phi^{-12}}{\sqrt{5}} \approx 0.001388875\dots,$$

while $1/720 = 0.001388888\dots$. The difference is the small residual

$$\varepsilon \approx -1.39 \times 10^{-8}.$$

Thus,

$$\frac{\phi^{12}}{\sqrt{5}} = 144 + \frac{1}{720} + \mathcal{O}(10^{-8}).$$

This identity connects the golden ratio, the perfect-square term $F_{12} = 144$ in the Fibonacci sequence, and the factorial $6! = 720$.

References

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